AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.081 / 0.0158543 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-22 20:00 UTC
Location	29.5203, -95.3925
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 741	29.01 to 100; mean 71.4753	42.21 at 2026-07-22 19:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 741	23.2222 to 39; mean 29.9344	37.1111 at 2026-07-22 19:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 885	0 to 360; mean 176.96	270 at 2026-07-22 19:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 909	0 to 8.74555; mean 2.9378	2.57222 at 2026-07-22 19:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 96	5915.17 to 5958.34; mean 5933.54	5939.78 at 2026-07-22 18:00Z; 2 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 96	61.3235 to 2017.14; mean 643.169	1888.34 at 2026-07-22 18:00Z; 2 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 96	29.7729 to 67.4723; mean 49.2769	55.6466 at 2026-07-22 18:00Z; 2 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 96	1003.61 to 1014.29; mean 1008.62	1008.05 at 2026-07-22 18:00Z; 2 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 96	17.875 to 24.7709; mean 21.2651	21.2281 at 2026-07-22 18:00Z; 2 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 96	-4.5306 to 4.0994; mean 0.8584	1.49198 at 2026-07-22 18:00Z; 2 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 96	-6.0398 to 5.07228; mean -0.2816	-1.68513 at 2026-07-22 18:00Z; 2 h before event
openaq	Ozone (ozone)	ppm	16 / 1073	-0.004 to 0.092; mean 0.0276	0.081 at 2026-07-22 20:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 179	3 to 98; mean 37.067	35 at 2026-07-22 20:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	74 / 2820	0 to 34.1; mean 9.5119	11 at 2026-07-22 20:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	4 / 289	0 to 100; mean 46.1384	84 at 2026-07-22 20:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	4 / 289	43 to 96; mean 73.308	48 at 2026-07-22 20:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	4 / 289	0 to 31.24; mean 0.3784	0 at 2026-07-22 20:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	4 / 289	1007 to 1015; mean 1010.56	1009 at 2026-07-22 20:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	4 / 289	24.14 to 40.07; mean 30.6529	39.28 at 2026-07-22 20:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	4 / 289	0 to 360; mean 188.149	7 at 2026-07-22 20:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	4 / 289	0.4 to 7.61; mean 2.4234	1.4 at 2026-07-22 20:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	68 / 4887	0 to 149.3; mean 11.6021	13 at 2026-07-22 20:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0233633 to 0.0312825; mean 0.0271	0.0311848 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000163343 to 0.000396712; mean 0.0002	0.000396712 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	2 / 2	5.0358e-05 to 5.87309e-05; mean 0.0001	5.0358e-05 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-2.41161e-06 to 9.61895e-05; mean 0	-2.41161e-06 at 2026-07-22 19:14Z; 45 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 45 min before event
tceq	Carbon monoxide (co)	ppm	3 / 195	0 to 0.5; mean 0.1436	0.1 at 2026-07-22 20:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 713	-2.6 to 44.3; mean 6.3403	4.1 at 2026-07-22 20:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 191	-1 to 1.6; mean -0.2063	-0.2 at 2026-07-22 20:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-21 06Z 94.7, 12Z 75.16, 18Z 40.47, 07-22 00Z 62.64, 06Z 80.64, 12Z 60.51, 18Z 42.71, 07-23 00Z 69.82, 06Z 75.28, 12Z 74.34, 18Z 85.9, 07-24 00Z 92.69, 06Z 87.69
asos	Air temperature (temperature)	07-21 06Z 24.76, 12Z 28.83, 18Z 35.31, 07-22 00Z 30.51, 06Z 26.85, 12Z 31.57, 18Z 37.7, 07-23 00Z 31.47, 06Z 29.26, 12Z 29.23, 18Z 28.1, 07-24 00Z 25.98, 06Z 25.89
asos	Wind direction (wind_direction)	07-21 06Z 172.7, 12Z 259.6, 18Z 227.4, 07-22 00Z 207.9, 06Z 240.1, 12Z 287.6, 18Z 217, 07-23 00Z 132.8, 06Z 44.21, 12Z 55.26, 18Z 159.2, 07-24 00Z 151.4, 06Z 93.64
asos	Wind speed (wind_speed)	07-21 06Z 1.909, 12Z 3.43, 18Z 2.408, 07-22 00Z 2.464, 06Z 2.917, 12Z 3.779, 18Z 2.691, 07-23 00Z 2.051, 06Z 3.073, 12Z 4.459, 18Z 2.751, 07-24 00Z 3.219, 06Z 2.105
noaa_gfs	500-hPa geopotential height (gh_500)	07-21 12Z 5917, 18Z 5932, 07-22 00Z 5929, 06Z 5935, 12Z 5926, 18Z 5939, 07-23 00Z 5928, 06Z 5942, 12Z 5926, 18Z 5933, 07-24 00Z 5939, 06Z 5956
noaa_gfs	Planetary boundary layer height (pbl_height)	07-21 12Z 192.2, 18Z 1172, 07-22 00Z 749.3, 06Z 121.6, 12Z 224, 18Z 1762, 07-23 00Z 501.4, 06Z 312.3, 12Z 307.4, 18Z 1336, 07-24 00Z 533.6, 06Z 505.9
noaa_gfs	Precipitable water (precipitable_water)	07-21 12Z 30.96, 18Z 34.41, 07-22 00Z 38.76, 06Z 45.3, 12Z 52.17, 18Z 55.56, 07-23 00Z 60.84, 06Z 48.38, 12Z 49.51, 18Z 61.99, 07-24 00Z 63.86, 06Z 49.56
noaa_gfs	Surface pressure (surface_pressure)	07-21 12Z 1011, 18Z 1010, 07-22 00Z 1008, 06Z 1008, 12Z 1009, 18Z 1008, 07-23 00Z 1005, 06Z 1008, 12Z 1008, 18Z 1007, 07-24 00Z 1008, 06Z 1013
noaa_gfs	850-hPa temperature (t_850)	07-21 12Z 22.55, 18Z 22.52, 07-22 00Z 21.96, 06Z 22.51, 12Z 21.66, 18Z 21.53, 07-23 00Z 23.86, 06Z 21.61, 12Z 20.08, 18Z 19, 07-24 00Z 19.79, 06Z 18.11
noaa_gfs	10-m eastward wind (u_10m)	07-21 12Z 2.608, 18Z 2.561, 07-22 00Z 1.44, 06Z 2.561, 12Z 3.114, 18Z 0.9357, 07-23 00Z 0.2144, 06Z -0.7155, 12Z -0.4267, 18Z -1.648, 07-24 00Z 0.8824, 06Z -1.225
noaa_gfs	10-m northward wind (v_10m)	07-21 12Z 0.3978, 18Z -0.5669, 07-22 00Z 0.2895, 06Z 1.408, 12Z -0.2864, 18Z -2.19, 07-23 00Z 1, 06Z -2.842, 12Z -3.384, 18Z -5.151, 07-24 00Z 4.518, 06Z 3.427
openaq	Ozone (ozone)	07-21 06Z 0.006483, 12Z 0.01149, 18Z 0.04328, 07-22 00Z 0.02801, 06Z 0.006478, 12Z 0.0157, 18Z 0.06599, 07-23 00Z 0.03987, 06Z 0.02748, 12Z 0.02719, 18Z 0.02741, 07-24 00Z 0.02114, 06Z 0.01411
openaq	PM10 (pm10)	07-21 06Z 38.75, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 30.56, 12Z 25.22, 18Z 16.39, 07-24 00Z 19.61, 06Z 38.22
openaq	PM2.5 (pm25)	07-21 06Z 7.237, 12Z 7.029, 18Z 5.838, 07-22 00Z 5.928, 06Z 5.902, 12Z 6.402, 18Z 9.251, 07-23 00Z 10.59, 06Z 17.95, 12Z 17.95, 18Z 8.838, 07-24 00Z 8.065, 06Z 11.89
openweather	Cloud cover (cloud_cover)	07-21 06Z 0, 12Z 1.435, 18Z 1.72, 07-22 00Z 3.792, 06Z 21.79, 12Z 11.79, 18Z 64.96, 07-23 00Z 79.22, 06Z 69.72, 12Z 98.5, 18Z 99.21, 07-24 00Z 94.54, 06Z 27.88
openweather	Relative humidity (humidity)	07-21 06Z 91.53, 12Z 80.04, 18Z 48.72, 07-22 00Z 62.04, 06Z 80.25, 12Z 70.88, 18Z 49.12, 07-23 00Z 71.3, 06Z 76.68, 12Z 77.42, 18Z 78, 07-24 00Z 93.5, 06Z 93.12
openweather	Precipitation rate (precipitation)	07-21 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.0075, 07-23 00Z 1.792, 06Z 0, 12Z 0, 18Z 0.485, 07-24 00Z 2.293, 06Z 0.1588
openweather	Surface pressure (pressure)	07-21 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1010, 12Z 1010, 18Z 1009, 07-24 00Z 1012, 06Z 1015
openweather	Air temperature (temperature)	07-21 06Z 25.35, 12Z 28.8, 18Z 36.47, 07-22 00Z 31.98, 06Z 27.36, 12Z 31.11, 18Z 38.98, 07-23 00Z 32.56, 06Z 29.94, 12Z 29.6, 18Z 29.95, 07-24 00Z 25.77, 06Z 25.3
openweather	Wind direction (wind_direction)	07-21 06Z 179.8, 12Z 240.1, 18Z 248.3, 07-22 00Z 228.5, 06Z 236.7, 12Z 255.9, 18Z 197.4, 07-23 00Z 231.8, 06Z 115.6, 12Z 53.75, 18Z 119, 07-24 00Z 161.1, 06Z 164.1
openweather	Wind speed (wind_speed)	07-21 06Z 2.384, 12Z 1.833, 18Z 1.504, 07-22 00Z 2.566, 06Z 1.875, 12Z 1.85, 18Z 2.249, 07-23 00Z 2.842, 06Z 2.296, 12Z 3.23, 18Z 4.159, 07-24 00Z 2.487, 06Z 1.914

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-21 06Z 9.729, 12Z 8.528, 18Z 7.807, 07-22 00Z 9.675, 06Z 6.904, 12Z 8.049, 18Z 11.18, 07-23 00Z 14.58, 06Z 22.7, 12Z 17.97, 18Z 10.07, 07-24 00Z 10.48, 06Z 13.97
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	CO column (s5p_co_column)	07-21 18Z 0.02656, 07-22 18Z 0.03123, 07-23 18Z 0.02345
sentinel5p	CO granules available (s5p_co_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-21 18Z 0.0002243, 07-22 18Z 0.0003967, 07-23 18Z 0.0001668
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06, 07-23 18Z 5.971e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
tceq	Carbon monoxide (co)	07-21 06Z 0.1333, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1333, 12Z 0.1333, 18Z 0.1611, 07-24 00Z 0.22, 06Z 0.1111
tceq	Nitrogen dioxide (no2)	07-21 06Z 5.767, 12Z 5.402, 18Z 4.761, 07-22 00Z 10.57, 06Z 6.367, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.105, 06Z 5.303, 12Z 6.175, 18Z 6.863, 07-24 00Z 7.503, 06Z 5.043
tceq	Sulfur dioxide (so2)	07-21 06Z -0.275, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3, 12Z -0.2333, 18Z -0.3278, 07-24 00Z -0.27, 06Z -0.3222

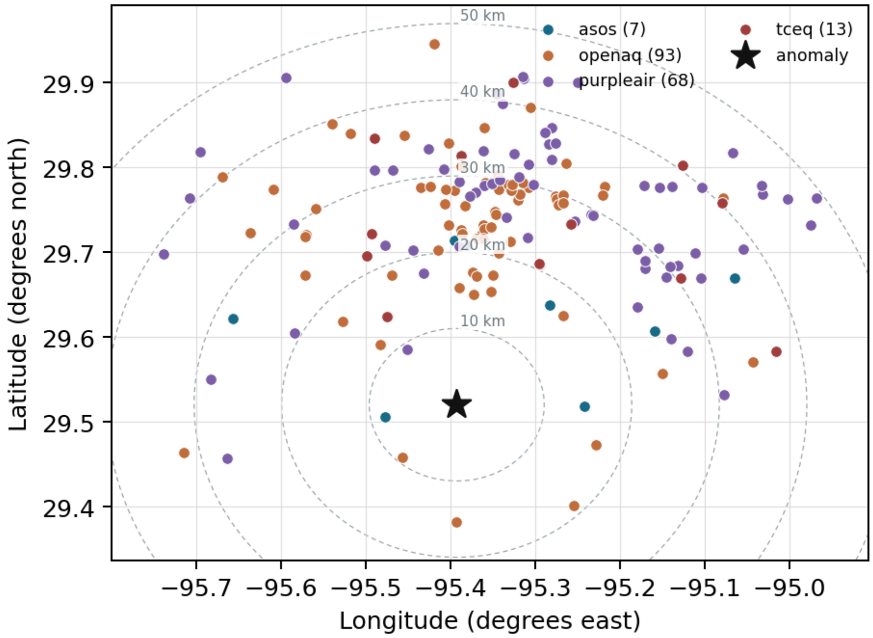
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

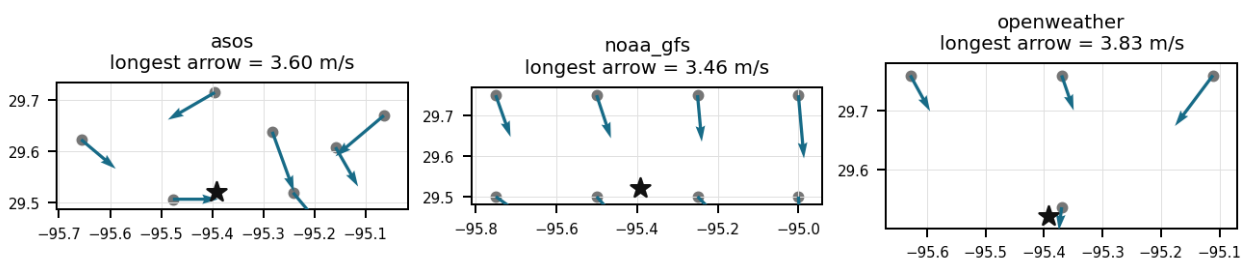
Source	Metric	Values
asos	Relative humidity (humidity)	AXH (8.318 km) 71.89, LVJ (14.592 km) 75.23, HOU (16.828 km) 71.27, MCJ (21.54 km) 67.16, EFD (24.595 km) 70.94, SGR (27.938 km) 73.91, T41 (35.81 km) 72.22
asos	Air temperature (temperature)	AXH (8.318 km) 29.46, LVJ (14.592 km) 29.32, HOU (16.828 km) 30.06, MCJ (21.54 km) 30.59, EFD (24.595 km) 30.24, SGR (27.938 km) 29.8, T41 (35.81 km) 29.97
asos	Wind direction (wind_direction)	AXH (8.318 km) 140.6, LVJ (14.592 km) 161.9, HOU (16.828 km) 180.5, MCJ (21.54 km) 198, EFD (24.595 km) 180, SGR (27.938 km) 196.1, T41 (35.81 km) 189.2
asos	Wind speed (wind_speed)	AXH (8.318 km) 1.59, LVJ (14.592 km) 1.968, HOU (16.828 km) 3.551, MCJ (21.54 km) 4.155, EFD (24.595 km) 3.43, SGR (27.938 km) 2.943, T41 (35.81 km) 3.194
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.50 (10.645 km) 5932, gfs:29.50,-95.25 (13.973 km) 5931, gfs:29.75,-95.50 (27.574 km) 5935, gfs:29.75,-95.25 (29.018 km) 5934, gfs:29.50,-95.75 (34.669 km) 5934, gfs:29.50,-95.00 (38.049 km) 5931, gfs:29.75,-95.75 (42.968 km) 5936, gfs:29.75,-95.00 (45.732 km) 5934
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.50 (10.645 km) 542.5, gfs:29.50,-95.25 (13.973 km) 556.4, gfs:29.75,-95.50 (27.574 km) 937.4, gfs:29.75,-95.25 (29.018 km) 863.1, gfs:29.50,-95.75 (34.669 km) 492.5, gfs:29.50,-95.00 (38.049 km) 538.8, gfs:29.75,-95.75 (42.968 km) 623.5, gfs:29.75,-95.00 (45.732 km) 591.1

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.50 (10.645 km) 48.96, gfs:29.50,-95.25 (13.973 km) 50.31, gfs:29.75,-95.50 (27.574 km) 48.36, gfs:29.75,-95.25 (29.018 km) 49.3, gfs:29.50,-95.75 (34.669 km) 48.21, gfs:29.50,-95.00 (38.049 km) 50.8, gfs:29.75,-95.75 (42.968 km) 48.09, gfs:29.75,-95.00 (45.732 km) 50.18
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.50 (10.645 km) 1008, gfs:29.50,-95.25 (13.973 km) 1009, gfs:29.75,-95.50 (27.574 km) 1008, gfs:29.75,-95.25 (29.018 km) 1009, gfs:29.50,-95.75 (34.669 km) 1008, gfs:29.50,-95.00 (38.049 km) 1010, gfs:29.75,-95.75 (42.968 km) 1007, gfs:29.75,-95.00 (45.732 km) 1010
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.50 (10.645 km) 21.25, gfs:29.50,-95.25 (13.973 km) 21.15, gfs:29.75,-95.50 (27.574 km) 21.34, gfs:29.75,-95.25 (29.018 km) 21.33, gfs:29.50,-95.75 (34.669 km) 21.25, gfs:29.50,-95.00 (38.049 km) 21.12, gfs:29.75,-95.75 (42.968 km) 21.39, gfs:29.75,-95.00 (45.732 km) 21.3
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.50 (10.645 km) 1.54, gfs:29.50,-95.25 (13.973 km) 0.7273, gfs:29.75,-95.50 (27.574 km) 0.5581, gfs:29.75,-95.25 (29.018 km) 0.4448, gfs:29.50,-95.75 (34.669 km) 1.302, gfs:29.50,-95.00 (38.049 km) 1.151, gfs:29.75,-95.75 (42.968 km) 0.7523, gfs:29.75,-95.00 (45.732 km) 0.3923
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.50 (10.645 km) -0.3001, gfs:29.50,-95.25 (13.973 km) -0.1709, gfs:29.75,-95.50 (27.574 km) -0.6542, gfs:29.75,-95.25 (29.018 km) -0.5792, gfs:29.50,-95.75 (34.669 km) -0.05589, gfs:29.50,-95.00 (38.049 km) 0.4249, gfs:29.75,-95.75 (42.968 km) -0.7659, gfs:29.75,-95.00 (45.732 km) -0.1517
openaq	Ozone (ozone)	3055 (0.0 km) 0.02893, 312 (13.969 km) 0.031, 320 (16.85 km) 0.02814, 325 (20.747 km) 0.009106, 311 (22.073 km) 0.03136, 3378 (27.1 km) 0.02797, 1319333 (28.091 km) 0.03251, 332 (30.475 km) 0.0338, +8 more
openaq	PM10 (pm10)	13380082 (27.1 km) 35.14, 15852419 (28.582 km) 30.75, 271 (30.475 km) 44.26
openaq	PM2.5 (pm25)	14152710 (9.263 km) 13.39, 13955814 (11.808 km) 5.256, 8967021 (14.63 km) 9.393, 15472101 (15.293 km) 6.134, 14140752 (15.341 km) 9.154, 11987924 (15.404 km) 7.454, 11638043 (16.705 km) 8.103, 14157975 (16.848 km) 10.05, +66 more
openweather	Cloud cover (cloud_cover)	grid:south (2.792 km) 50.92, grid:center (26.788 km) 45.77, grid:west (35.114 km) 43.51, grid:east (38.11 km) 44.36
openweather	Relative humidity (humidity)	grid:south (2.792 km) 73.33, grid:center (26.788 km) 71.1, grid:west (35.114 km) 71.54, grid:east (38.11 km) 77.29
openweather	Precipitation rate (precipitation)	grid:south (2.792 km) 0.404, grid:center (26.788 km) 0.2063, grid:west (35.114 km) 0.6304, grid:east (38.11 km) 0.2751
openweather	Surface pressure (pressure)	grid:south (2.792 km) 1011, grid:center (26.788 km) 1010, grid:west (35.114 km) 1011, grid:east (38.11 km) 1011
openweather	Air temperature (temperature)	grid:south (2.792 km) 30.34, grid:center (26.788 km) 31.24, grid:west (35.114 km) 30.9, grid:east (38.11 km) 30.11
openweather	Wind direction (wind_direction)	grid:south (2.792 km) 187.2, grid:center (26.788 km) 195.2, grid:west (35.114 km) 187.8, grid:east (38.11 km) 182.4
openweather	Wind speed (wind_speed)	grid:south (2.792 km) 2.459, grid:center (26.788 km) 2.353, grid:west (35.114 km) 1.646, grid:east (38.11 km) 3.237
purpleair	PM2.5 (pm25)	161003 (9.186 km) 6.149, 166451 (17.628 km) 11.9, 293735 (20.756 km) 12.18, 276512 (20.799 km) 12.97, 46237 (20.895 km) 0.3958, 293725 (20.922 km) 12.04, 186315 (22.489 km) 12.07, 6752 (23.328 km) 8.088, +60 more
tceq	Carbon monoxide (co)	482011035 (27.078 km) 0.1672, 482011039 (30.456 km) 0.09538, 482011052 (32.704 km) 0.1682
tceq	Nitrogen dioxide (no2)	480391004 (0.017 km) 4.933, 482010416 (20.738 km) 6.53, 482010055 (22.069 km) 3.847, 482011066 (24.389 km) 13.49, 482011035 (27.078 km) 7.057, 482011039 (30.456 km) 7.494, 482011052 (32.704 km) 4.124, 482010047 (36.131 km) 5.709, +4 more
tceq	Sulfur dioxide (so2)	482010051 (13.984 km) -0.4778, 482011035 (27.078 km) -0.4794, 482011039 (30.456 km) 0.3215

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

Precursor evidence favors VOC-rich and somewhat aged urban-industrial photochemistry more than a fresh local combustion plume because formaldehyde and carbon monoxide columns were elevated near the event while local surface NO2 was only 4.1 ppb and local SO2 was near zero at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

Coinciding wit h th e ozon e anomal y on July 22, 2026, ASOS station s and OpenWeather quer y point s recorde d elevate d surface temperature s exceeding 37 degC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Houston meteorology on 2026-07-22 20:00Z strongly favored rapid afternoon ozone production because temperatures were near 37–39 C, relative humidity was about 42–48%, winds were light at roughly 1.4–2.6 m/s, and boundary-layer height was very deep at about 1888 m.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

A deep and well-mixed boundary layer supports efficient regional accumulation and vertical mixing of ozone and precursors because modeled planetary boundary layer height near the anomaly was about 1888 meters at 18Z on 2026-07-22, far above the period mean of about 643 meters.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

Extreme afternoon heat, low humidity, and a deep boundary layer on 2026-07-22 likely accelerated photochemical ozone production in Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

Light-to-moderate and variable low-level winds on 2026-07-22 likely allowed locally emitted urban-industrial precursor plumes to remain concentrated enough to produce a sharp ozone peak in central Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

Unusual wind patterns were observed, including a sudden change in direction and speed, which may have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The anomaly occurred on July 22, 2026, at around 20:01:52 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The Houston ozone anomaly is most likely explained by secondary photochemical ozone in an aged urban-industrial plume rather than by a fresh local combustion or sulfur emission event because ozone reached 0.081 ppm while nearby NO_2 was only 4.1 ppb and nearby SO_2 was about -0.2 ppb at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

The location of the anomaly is approximately 2.792 km south of the query point.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

The air quality anomaly in Houston, Texas is not caused by high levels of NO_2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

The air quality anomaly in Houston, Texas on July 22nd was caused by a combination of factors including high PM2.5 levels and unusual wind patterns.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The openaq ozone network mean for 2026-07-22 18Z was 0.06599 ppm, which was higher than the 6-hour means at 2026-07-22 12Z (0.0157 ppm) and 2026-07-23 00Z (0.03987 ppm).

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air quality anomaly in Houston, Texas is not caused by high levels of SO_2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The openaq ozone monitor at 29.520, -95.392 measured 0.081 ppm at 2026-07-22T20:00:00+00:00, compared with an expected value of 0.01585430463576159 ppm, corresponding to a z-score of 5.690684734798128 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

Sentinel-5P precursor indicators on 2026-07-22 showed elevated formaldehyde and carbon monoxide near Houston, which is consistent with a VOC-reactive aged plume contributing to ozone formation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

Peak atmospheric concentrations of formaldehyde detected by Sentinel-5P on July 22, 2026, supplied reactive organic volatile precursors that accelerated ozone formation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

An OpenAQ monitoring station in Houston recorded a severe surface ozone concentration anomaly of 0.081 ppm at 20:00 UTC on July 22, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

Extreme afternoon heat and dryness strongly support rapid ozone photochemistry in central Houston because air temperature near the anomaly reached about 37.1 to 39.3 degrees Celsius while relative humidity near the anomaly was only about 42 to 48 percent at the event time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

High PM2.5 levels were detected in the area, with values exceeding 13 ug/m3 at multiple sensors.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

TCEQ ground monitors registered average NO2 concentrations of approximately 6.3 ppb around the time of the anomaly, suggesting precursor balance was strongly influenced by organic reactivity.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

A sever e ozon e anomal y was detected at coordinate s (29.520, -95.392) on 2026-07-22T20:00:00+00:00 wit h a value of 0.081 ppm and a z-scor e of 5.69.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

Th e 6-hour mean ozon e concentration for th e region peake d at 0.06599 ppm during th e 18Z period on July 22, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

High daytime surface temperatures exceeding 37 to 39 degrees Celsius on July 22, 2026, accelerated photochemical reaction rates to produce elevated ground-level ozone in Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

Sentinel-5P satellite observations revealed a peak formaldehyde column density of 0.000397 mol/m2 on July 22, confirming an influx of reactive volatile organic precursors during the ozone event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

Elevated formaldehyde columns on 2026-07-22 indicate abundant reactive VOC precursors that likely enhanced ozone formation over the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

The anomaly is a result of a combination of factors including temperature inversions, stagnant air masses, and high levels of ozone precursors in the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

Planetary boundary layer expansion reaching approximately 1800 meters during midday provided strong vertical mixing conditions that distributed ozone precursors across the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Surface temperatures reaching over 37 degrees Celsius recorded by ASOS and OpenWeather, along with a GFS planetary boundary layer height reaching 1888 meters, created thermal and convective conditions favorable for ozone production.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

Near the anomaly time and location, local observations showed afternoon heat, dryness, and light winds, including 39.28 degC temperature, 48 percent humidity, and 1.4 m/s wind speed at 2026-07-22T20:01:52+00:00 in openweather, and 37.11 degC temperature, 42.21 percent humidity, and 2.57222 m/s wind speed at 2026-07-22T19:55:00+00:00 in asos.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

Surface meteorology on July 22, 2026, exhibited intense daytime heating with temperatures reaching 37°C to 39°C alongside atmospheric planetary boundary layer heights exceeding 1800 meters.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 13.0 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

Sentinel-5P satellite measurements detected an elevated formaldehyde column concentration of 0.000397 mol/m² over the Houston area at 19:14 UTC on July 22, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.